O-C-101a
21 JULY 1949
SUPERSEDING
Fed. Spec. O-C-101
JULY 22, 1930

FEDERAL SPECIFICATION

CALCIUM CARBIDE

This specification was approved by the Director, Bureau of Federal Supply, for the use of all departments and establishments of the Government.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — Calcium carbide covered by this specification shall be suitable for the production of acetylene.

1.2 Grade and sizes.—Calcium carbide covered by this specification shall be of but one grade and shall be of the size designations listed in table I, as specified. (See 3.2.)

2. APPLICABLE SPECIFICATIONS

2.1 Specifications. — There are no other specifications applicable to this specification except those referenced in section 7 by individual departments.

(Copies of Federal specifications and Federal Specifications Index may be obtained upon application, accompanied by postal note, money order, coupon, or cash to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

3. REQUIREMENTS

3.1 Material.—Calcium carbide shall be the commercial grade of material normally containing small percentages of silicide of iron, uncombined carbon, and unconverted raw materials. It shall be free from metal particles and other foreign materials except as specified.

3.2 The calcium carbide shall conform to the requirements shown in table I for the respective size designations.

TABLE I.

Size designation	Pass through sieve with square openings[1]	Retained on sieve with square openings[1]	Minimum average volume of acetylene gas[2] evolved per pound at 60°F. (15.56° C.) and 30 inches (760 mm.) barometric pressure
	Inches	*Inches*	*Cubic feet per pound*
Lump	4.24	1.50	4.5
Egg	2.00	.375	4.5
Nut	1.06	.250	4.5
½ by ¼[3]	([4])	([4])	4.5
¼ by 1/12	([5])	([5])	4.5
Rice	0.132	0.0331	4.3
14ND	0.0661	0.0165	4.3

[1] For all sizes except the ½ by ¼ size and the ¼ by 1/12 size, not more than 5 percent, by weight of calcium carbide other than the nominal size shall be present.

[2] The evolved acetylene gas produced from the carbide shall contain not more than 0.05 percent, by volume, of phosphine (PH_3).

[3] The ½ by ¼ size designation is also known as Miner's Lamp, and Miner's Special.

[4] The ½ by ¼ size carbide shall conform to the following granulation:

Through sieve with square openings: *Inches*	*Percent*
0.500	100 (minimum)
0.375	95 (maximum)
0.250	50 (maximum)
0.0937	5 (maximum)

[5] The ¼ by 1/12 size carbide shall conform to the following granulation:

Through sieve with square openings: *Inches*	*Percent*
0.265	100 (minimum)
0.0937	25 (maximum)
0.0661	7 (maximum)
0.0555	2 (maximum)

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—From one of each lot of 10 drums or fraction thereof, a representative sample of not less than 0.5 pound shall be drawn. These samples shall be mixed to constitute a composite sample. For large lots, a composite sample shall be prepared for each 100 drums or fraction thereof, and the average of the gas yields of these composite samples shall be taken as the gas yield of the lot. The composite sample shall be placed in an air-tight container, marked with the proper identification, and forwarded to the laboratory for test.

4.2 Inspection.—The calcium carbide shall be inspected for compliance with the size designation requirements of 3.2.

4.3 Chemical tests.

4.3.1 *Gas yield.*

4.3.1.1 *Preparation of sample.*—For material sized ½ by ¼ or smaller, the entire composite sample shall be put through a riffle and cut down to approximately 1 pound. This shall be done as rapidly as possible with every effort to minimize exposure of the carbide to air. This prepared sample shall be preserved in an air-tight container until tested. For material coarser than ½ by ¼, the entire sample shall be quickly put through a jaw crusher (6-inch by 3-inch Bacon crusher or equivalent) and a set of riffles to cut it down to approximately 1 pound. Samples of carbide coarser than 1¼ inches by ⅜ inch may be crushed initially in a 10-inch by 7-inch or larger jaw crusher. The crusher shall be operated to break the sample so that as much as possible is left as lumps approximately ¼ inch in diameter and a minimum of fine dust is produced. During the crushing and riffling, exposure of the sample to air shall be kept at a minimum. The prepared sample shall be mixed and stored in an air-tight container until tested.

4.3.1.2 *Apparatus.*

(1) *Generating flask.* — 500-milliliter round-bottom, three-neck, angle-type flask, with ground joints (Scientific Glass Apparatus Co., Catalog No. 9157 or equivalent). Flask shall be cooled by water jets directed against it from a spray ring. Flask shall be provided with one glass stopper. A shield of wire screen glass or safety glass shall be provided to protect the operator during the test.

(2) *Carbide feeder.* — 50-milliliter Erlenmeyer flask connected to ground joint with approximately 8 inches of ½-inch diameter transparent plastic tubing closed with a pinch clamp.

(3) *Cooler.* — Bulb-type condenser with ground joint and 20-centimeter jacket length.

(4) *Scrubber.*—500-milliliter flask (Scientific Glass Apparatus Co., Catalog No. 9146 or equivalent), with stopper and a ground joint containing a thermometer mounted in an open well by means of a rubber sleeve. Glass stopper containing a sealed-in delivery tube with its tip approximately ¾ inch from the bottom of the flask. Right-angle exit tube extending just through the stopper. Flask filled with glass wool wet with 50 milliliters of water. Water shall be added to replace evaporation.

(5) *Wet test meter.* — (Sargent-Central Scientific Catalog No. 20740 or equivalent.) This test meter should be calibrated frequently with a meter test bottle with a National Bureau of Standards certificate or by displacement with water as follows:

Provide a 12-liter aspirator bottle with the top closed with a tight-fitting rubber stopper containing a ⅜-inch glass elbow and with the bottom end closed with a screw clamp on ⅜-inch rubber tubing. The end of the rubber tubing is connected to a ¼-inch glass elbow. Connect the ⅜-inch glass elbow to the outlet of the wet meter with ¼-inch rubber tubing containing a three-way stopcock.

Fill the aspirator bottle with water of the same temperature as the meter, open the screw clamp to fill the ¼-inch glass elbow, then with the three-way stopcock turned so

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that the bottle is open to the air, insert the stopper firmly in the neck of the bottle. Check the system carefully for air leaks, then turn the stopcock to connect the meter to the aspirator bottle, read the meter, and allow the water to run into a dry 2-liter flask until it is filled to the mark. Run at a maximum of 0.06 cubic foot per minute. Repeat with a second dry flask until at least 10 liters have passed through the meter. Read the meter. Convert liters to cubic feet using a multiplier of 0.035315. Divide the result by the difference in meter readings to obtain the correction factor. Before calibrating the meters, the manufacturer's instructions should be read and the meter filled with water exactly to the indicator mark reading the bottom of the meniscus. If there is play in the meter needle, read the middle of the play.

(6) *Barometer.*— (Eimer and Amend Catalog No. 2–380 or equivalent.)

4.3.1.3 *Procedure.*—To insure accurate determinations of gas yield, certain precautions are necessary. The temperature of the laboratory where the test is being made should not vary widely, and the meter temperature shall not vary more than 2° F. during a determination. The gas meter shall be operated at constant speed as these meters tend to register high at slower speeds than the one specified, and low at higher speeds. The meters shall be accurately leveled. The meter and scrubber shall be saturated with acetylene. The temperature in the generating flask shall be sufficiently low to prevent polymerization of the acetylene by local overheating. This is accomplished by means of a steady flow of cold water through the spray ring. The assembled apparatus shall be gastight.

After the apparatus has been assembled, place 250 milliliters of water in the generating flask, start the water flowing through the jacket of the cooler and the spray ring, and then saturate the system with acetylene as follows: Attach a feeder flask containing approximately 50 grams of carbide (not the sample being tested). Close the generator by inserting the glass stopper, and start slowly feeding the carbide into the water to attain a moderate rate of generation (0.05 to 0.06 cubic feet per minute). When the carbide feed is complete, shake the generating flask to wet any lumps. When the gas generation ceases, disconnect the feeder flask and discard the sludge in the generating flask. Place 250 milliliters of water in the generating flask and again feed in approximately 50 grams of carbide. Then weigh out two samples of approximately 50 grams each to the nearest 0.2 gram and transfer to dry feeder flasks. Treat these two portions in the same manner to insure that the system is thoroughly saturated with acetylene. The third and fourth corrected meter readings should agree to 0.02 cubic feet. The corrected meter reading for a 50-gram sample is obtained from the following relation:

$$\text{Corrected meter reading} = \frac{50 \times \text{meter reading in cubic feet}}{\text{grams of sample}}$$

Weigh approximately 50 grams (to the nearest 0.02 gm.) of calcium carbide representing the sample being tested and transfer to a dry feeder flask, closing the flask with a pinch clamp. When the meter hand ceases to turn from the last preliminary generation, disconnect the feeder flask, wash out the generator flask, and add 250 milliliters of water. Replace the flask, connect the feeder flask, replace the glass stopper, and after reading the meter, open the screw clamp to feed carbide into the water. Feed in the carbide at such a rate that the meter registers 0.05 to 0.06 cubic feet per minute. Take manometer and exit gas temperature readings frequently and record with the barometer reading. The temperature in the scrubber shall not differ by more than 5° F. from that of the meter. When all the carbide has been fed into the water, shake the flask. After the meter hand ceases to turn, record the final meter reading. Obtain the correction factor to reduce the indicated gas volume to moisture-saturated acetylene at 60° F. and 30 inches of mercury pressure as follows:

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$$\text{Correction factor} = \frac{(P-E)\ (459.6 + 60)}{(459.6 + T)\ (30.0 - 0.5216)}$$

where:

P = barometric pressure.
E = vapor pressure of water in inches of mercury at T° F.
T = temperature of acetylene as metered in °F.
459.6 = absolute temperature at 0° F.
0.5216 = inches of mercury pressure of water vapor at 60° F.

Calculate the gas yield as follows:

$$\text{Gas yield, cubic feet per pound} = \frac{\text{meter reading x correction factor x 453.6}}{\textit{grams of sample}}$$

The average of three 50 gram gas yield determinations shall be taken as the average gas yield of each of the prepared samples, provided that the difference between the highest and lowest values is less than 0.2 cubic feet per pound.

4.3.2 *Phosphine content of evolved gas.*

4.3.2.1 *Apparatus for generating acetylene.*—

(1) *500-milliliter generating flask.*

(2) *No. 6 two-hole stopper.*

(3) *500-milliliter dropping funnel.*

(4) *Right-angle glass tube with stopcock.*

(5) *Rubber tubing.*

(6) *Sulfur determination apparatus (absorber and spray trap only).* — (Corning Catalog No. 37723 or equivalent).

(7) *Glass trap.*—(Eimer and Amend Catalog No. 13–557 or equivalent.)

(8) *Funnel.*—7 inches in diameter.

(9) *Wet test meter.*

4.3.2.2 *Reagents.*—

(1) *Sodium chloride solution.*—Saturated at room temperature.

(2) *Iodine solution.* — Approximately 0.1N.

(3) *Concentrated nitric acid.*

(4) *Nitric acid wash solution.*—(1:59.)

(5) *Concentrated ammonium hydroxide.*

(6) *Stopcock grease.* — (Dow-Corning or equivalent.)

(7) *Potassium nitrate.* — 1-percent solution.

(8) *Phenolphthalein indicator solution.*

(9) *Ammonium molybdate solution.* — Weigh 100 grams of 85-percent molybdic acid, place in a beaker, and mix thoroughly with 240 milliliters of distilled water. Add 140 milliliters of concentrated NH_4OH, filter, and add 60 milliliters of concentrated HNO_3. Designate this solution No. 1. Mix 400 milliliters of concentrated HNO_3 and 960 milliliters of distilled water. Cool the nitric acid solution and solution No. 1 to room temperature. Add solution No. 1 to the nitric acid solution while stirring constantly. Add 0.01 gram of ammonium phosphate dissolved in 10 milliliters of distilled water and let stand for 24 hours before using. Use only the clear supernatant liquid and store in a cool place in a glass-stoppered bottle.

(10) *Standard sodium hydroxide.*—0.1N. Standardize by titrating against potassium acid phthalate using phenolphthalein indicator.

(11) *Standard nitric acid.*—0.1N. Standardize against the standard NaOH using phenolphthalein indicator and make it equivalent to the alkali by dilution or additions of acid.

4.3.2.3 *Generation.*—An accurately weighed sample of calcium carbide, between 50 to 75 grams, is placed in the generating flask. The flask is closed with the rubber stopper, through which is inserted the dropping funnel and outlet tube. The outlet tube is connected, gas-tight, to the glass trap which is connected to the absorber. The absorber, which should be cleaned with warm concentrated HNO_3 before using, is filled

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with 100 milliliters of 0.1N iodine solution. The dropping funnel is filled with approximately 300 milliliters of the NaCl solution. Rubber tubing is connected from the spray trap through a wet test meter to an outside vent. Gentle pressure is applied to the NaCl solution to overcome the back pressure of the generator by means of cylinder nitrogen. The generator flask is kept cool by cold water flowing over it into a funnel underneath. A screw clamp on the funnel drain tube will serve to regulate the amount of contact between water and flask. The NaCl solution is added dropwise, 6 to 7 drops per minute, to the flask containing the calcium carbide. After approximately 3 hours, the carbide will be completely decomposed. When the evolution of gas has ceased, the remainder of the NaCl solution in the dropping funnel is permitted to flow onto the carbide sludge, and gas from the nitrogen cylinder is used to sweep the last traces of acetylene from the apparatus. The solution in the absorber is blown into the smaller absorber bulb before it is transferred to a beaker. The absorber and spray trap are rinsed three times with distilled water and the rinsings added to the original solution.

4.3.2.4 *Analysis.*—To the absorbing solution which has been transferred to a 500-milliliter widemouth Erlenmeyer flask, 35 milliliters of concentrated HNO_3 are added. The solution is boiled on a hot plate until all iodine has evolved. The solution is cooled, then neutralized with NH_4OH, using litmus or Hydrion wide-range test paper as the indicator. To the neutralized solution, the volume of which should be a minimum of 100 milliliters, add 5 milliliters HNO_3 per 100 milliliters of solution. The acidified solution is then heated to 40° to 45° C. before stirring in 75 milliliters of the ammonium molybdate solution. Stopper the flask and shake for 10 minutes, keeping the solution at approximately 45° C. Allow the precipitate to settle for 1 hour. The yellow precipitate is filtered off, washed once or twice with 1:50 HNO_3 solution, then with 1-percent KNO_3 solution until the filtrate is neutral to litmus paper. Filtration through asbestos is preferable although a close filter paper may be used. The precipitate is transferred to a 400-milliliter beaker and approximately 100 milliliters of CO_2 free water is added. An excess of 0.1N NaOH is then added from a buret, and enough time is allowed to insure that all the precipitate has been dissolved. The excess NaOH is titrated with 0.1N HNO_3. A blank is run on the reagents alone.

From the reaction:

$$2(NH_4)_3PO_4.12MoO_3 + 46NaOH = 2(NH_4)_2\ HPO_4 + (NH_4)_2MoO_4 + 23Na_2MoO_4 + 22H_2O$$

it is evident that 23 equivalents of NaOH are equivalent to one equivalent of phosphorus and therefore phosphine. Therefore, 1 milliliter of NaOH equals 0.9737 milliliter PH_3.

$$\text{Volume of dry } PH_3 \text{ (at S.T.P.) in milliliter per gram of sample taken} = \frac{[VN - vn - (V_1N - v_1n)] \times 0.9737}{W}$$

where:

V = milliliter NaOH used in reacting sample.
N = normality of NaOH.
v = milliliter HNO_3 used in back titration of sample.
n = normality of HNO_3.
V_1 = milliliter NaOH used in blank.
v_1 = milliliter HNO_3 used in back titration of blank.
W = gram of sample taken.

Calculate the percent, by volume, of phosphine in the evolved acetylene as follows:

$$\text{Percent phosphine} = \frac{v}{V} \times 100$$

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where:

v = milliliter of phosphine (at S.T.P.) per gram of sample taken.
V = milliliter of acetylene (at S.T.P.) evolved per gram of carbide obtained as specified in 4.3.1.3 and converted from cubic feet per pound.

Note.—The precision of the method may be checked by adding 100 milliliters of the 0.1N iodine solution to 35 milliliters of HNO_3. Boil until all iodine is expelled, then dissolve 2 grams of National Bureau of Standards sample 8e in the solution and determine the phosphorus content of the steel.

5. PREPARATION FOR DELIVERY

5.1 **Packing.** — Unless otherwise specified, calcium carbide shall be packed in steel drums containing 100 pounds net. The drums shall be of sufficient strength to insure handling without rupture. They shall be so constructed as to be water- and air-tight without the use of solder, except at such points as it may be necessary to use solder in order to close small crevices which cannot otherwise be sealed. Solder shall not be used on joints in such a manner that fire would disrupt the package. Drums shall be painted with high-grade rust-proof paint.

5.2 **Marking.** — Drums shall be clearly marked with the name of the contractor of the carbide, size of the carbide, and the wording "Calcium Carbide — Dangerous If Not Kept Dry". Each drum shall bear a tag or be marked so as to show the number of the contract or order.

6. NOTES

6.1 **Ordering data.**—Purchasers should specify the size designation of calcium carbide required (see table I) in accordance with the manufacturer's recommendations for the particular equipment being operated.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and other publications, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 **Army, Air Force, and Navy.**

7.2.1 *Applicable specifications and other publications.*—

National Military Establishment Specification:

JAN-D-195 — Drums, Steel, Calcium Carbide. (100-Pound Capacity.)

U. S. Army Specifications:

94-40645—Marking; Exterior, Domestic and Export Shipment, by Contractors.[1]

100-2—Standard Specification for Marking Shipments by Contractors.[2]

Navy Department Specification:

General Specifications for Inspection of Material.[3]

Bureau of Supplies and Accounts Publication:

Navy Shipment Marking Handbook.[3]

(Copies of specifications required for Army purchases may be obtained upon application to the procuring agency. Both the title and identifying symbol or number should be stipulated when requesting copies.

(Copies of specifications required for Air Force purchases may be obtained from the Commanding General, Air Matériel Command, Wright-Patterson Air Force Base, Dayton, Ohio.

[1] Applicable to Air Force purchases only.
[2] Applicable to Army purchases only.
[3] Applicable to Navy purchases only.

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(Copies of Federal and National Military Establishment specifications required for Navy purchases, Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol or number should be stipulated when requesting copies of specifications.)

7.2.2 *Packing for domestic and overseas shipment.*—Calcium carbide shall be packed for domestic and overseas shipment in steel drums conforming to NME Specification JAN–D–195.

7.2.3 *Marking.* — Unless otherwise specified, marking of shipping containers for the Army shall be in accordance with U. S. Army Specification 100–2; for the Air Force, in accordance with U. S. Army Specification 94–40645; and for the Navy, in accordance with the Navy Shipment Marking Handbook.

NAVY: SHS

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEM PROCURED | DOLLAR AMOUNT
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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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